

Hierarchical Transformer Preconditioning for Real-Time Simulation

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We outline research on hierarchical transformer-based preconditioners for linear systems that arise in time-stepping fluid simulation, with the goal of real-time or interactive performance after amortized training. This submission template follows ACM acmtog instructions for SIGGRAPH Asia Technical Papers; replace this abstract with the final text.

CCS Concepts: • **Computing methodologies** → **Physical simulation**.

Additional Key Words and Phrases: preconditioning, iterative solvers, neural operators, fluid simulation, real-time graphics

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1 Introduction

Real-time fluid simulation often reduces to repeatedly solving large, sparse symmetric positive definite linear systems, for example in pressure projection steps [Lyu et al. 2026]. Classical multigrid and Krylov methods remain strong baselines [Luz et al. 2020], while learning-based approaches have targeted preconditioners and fast surrogate operators [Chen 2025; Li et al. 2023; Xu et al. 2025; Yang et al. 2025]. Hierarchical and low-rank structure appears both in classical fast algorithms and in recent neural architectures for Green’s operators and PDE maps [Fan et al. 2019; Fognini et al. 2025; Sittoni et al. 2026; Yang et al. 2026]. Differentiable simulation and inverse problems further motivate fast linear algebra inside optimization loops [Hartland et al. 2023; Li et al. 2024, 2025; Ma et al. 2023]. We develop a learning-based sparse approximate inverse (SPAI) preconditioner built from a recursive hierarchical transformer (LEAFONLY architecture) for use inside preconditioned conjugate gradient (PCG) solves.

- **Architecture.** Recursive hierarchical transformer preconditioner with block tokens, hierarchical attention, GCN layers, and highway connections (Section 4).
- **Training.** Self-supervised objective based on Hutchinson-type estimators (cosine vs. L_2 variants) for operator quality without paired ground-truth solves.
- **Performance.** GPU-friendly SPAI structure avoiding triangular solves; target real-time total solve time with reported iteration counts and residuals vs. baselines (Jacobi, IC, AMGX, unpreconditioned CG).

2 Related Work

- 2.1 Classical Iterative Solvers and Preconditioners
- 2.2 Neural Operators and Machine Learning for Physics
- 2.3 Learning-Based Preconditioners

3 Background

- 3.1 The Conjugate Gradient Method
- 3.2 $\mathcal{H}$ -Matrix Approximation
- 3.3 Hutchinson Trace Estimation

4 Method: Recursive Hierarchical Transformer Preconditioner

- 4.1 Feature Representation
- 4.2 Network Architecture
- 4.3 Operator Construction

5 Training Methodology

- 5.1 Objective Function
- 5.2 Training Dynamics

6 Experiments and Results

- 6.1 Performance and Speedup
- 6.2 Generalization and Robustness
 - 6.2.1 Scale and Problem Types.
 - 6.2.2 Contrast Ratios.
 - 6.2.3 Precision Limits.

6.3 Hardware Efficiency

6.4 Ablation Studies

7 Limitations and Future Work

8 Conclusion

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